

KineTrace

Detecting Early Motor Irregularities Using Wearable Motion Sensors and Signal Processing.

Overview

Purpose

The goal of this project is to utilize motion sensor data from everyday consumer devices to determine measurable biomechanical patterns that may correlate with early motor irregularities. More specifically, I want to use frequency-domain analysis (FDA) and smoothness metrics extracted from smartwatch IMU data in order to distinguish normal movement from tremor-like or unstable movement patterns. These methods have been used before in order to monitor people already diagnosed with disorders that signal motor dysfunction such as Parkinson’s disease, however I aim to identify subtle losses in motor smoothness and stability that appear before obvious symptoms are visible by identifying micro-irregularities and hidden motion instability in human movement. Furthermore, in order to strengthen my dataset, I will use both public datasets and create my own data. Below I have included a small table to quickly summarize the key aspects of my research.

Method	Purpose
Frequency Domain Analysis	I will be using FDA in order to detect hidden oscillations/micro-tremors.
Jerk Analysis	I will analyze jerk to determine movement smoothness.
Activity Recognition	I will distinguish tremors from normal movement.
Hybrid Dataset System	I will use both public datasets and self-collected data.

Community Impact and Relevance

In addition to detecting subtle motor irregularities through wearable motion sensors, KineTrace will also examine how culture, community, and lifestyle factors may influence the risk of neurological motor diseases, particularly within elderly populations. Using the KineTrace Stability Index (KSI) which is going to be the end goal of my research, this project aims to identify early signs of reduced movement smoothness, instability, or tremor-like behavior before major symptoms become visible. My research will explore how factors such as physical activity, social interaction, participation in community programs, mobility independence, and access to healthcare may affect motor stability

over time. By combining signal processing with community health analysis, KineTrace expands beyond a technical engineering project and investigates how accessible wearable technology could support earlier detection, preventative care, and improved quality of life for aging populations.

Why Frequency-Domain Analysis?

Frequency-domain analysis or FDA is a method of analyzing signals by breaking them down into their individual frequency components rather than viewing how they change over time. FDA will be used in this project in order to analyze motion data collected from the accelerometers and gyroscopes already present inside smartwatches. Using FDA will be more useful than viewing how data changes over time for this project since irregular motor movement and tremor-like behavior often occurs as repeating oscillations at specific frequencies that are difficult to detect from simple motion graphs. In order to utilize FDA, I will be using Fast Fourier Transforms (FFTs) in order to convert signals from the time domain (amplitude vs. time) into the frequency domain (amplitude vs. frequency). This will allow me to identify dominant motion frequencies, measure oscillation intensity, and compare stable movement patterns against irregular ones quantitatively.

What is Smartwatch IMU Data?

Smartwatch inertial measurement unit (IMU) data is raw, high-frequency motion data that is collected by the smartwatch sensors. The sensors that I am going to be utilizing for this project are the accelerometer and gyroscope. These sensors measure linear acceleration and rotational movement in the three-dimensional space and continuously record motion patterns. This means I can convert movement into numerical data that can later be analyzed using frequency analysis. Since it is hard to find an application that is able to collect data from smartwatch sensors, I will be using these same sensors but on a smartphone since both use MEMS (Micro-Electro-Mechanical-Systems) technology to detect motion and orientation.

Research Question

Can smartwatch IMU data and signal-processing techniques detect subtle movement irregularities early before obvious motor dysfunction is visible?

Process

End Goal

Create a system that does the following:

1. Collects motion data from wearable sensors.
2. Analyzes movement using physics.
3. Measures motion smoothness and stability.
4. Distinguishes stable movement from irregular movement.

5. Create a web application to allow the public to use the system that is built on the algorithm created through the analysis of both collected data and public datasets

The final result of this will be the KineTrace Stability Index (KSI) which will be a numerical score that represents movement smoothness, motion stability, and oscillatory irregularity. I believe that this score could help people detect motor dysfunction early before it becomes a significant problem and take action in order to prevent medical emergencies.

Resources

Tool	Purpose
Smartphone	Collecting data using sensors.
Sensor Logger App	Recording data and exporting it as a table.
Python	Data analysis and computations.
VS Code	Environment for programming.
NumPy	Library for computations using Python.
pandas	Library for data analysis using Python.
matplotlib	Library for creating graphs in Python.
SciPy	Library for scientific computing in Python.
scikit-learn	Library for machine learning in Python.

Data Collection

Parameter	Specification	Benchmark Alignment
Sampling Rate	50 Hz (20ms sampling window)	Matches MotionSense & UCI HAR
Sensors	3-Axis Accelerometer (ax, ay, az) + 3-Axis Gyroscope (gx, gy, gz)	6-Axis IMU Template
Placement	Front Right Pants Pocket (Screen facing thigh, top pointing down)	MotionSense Standard
Activity Classes	std, sit, wlk, jog, ups, dws	Standard HAR 6-Class Taxonomy

Time Per Subject

3.5 Minutes Total

Assembly Line Execution

Phase 1: Hardware & App Setup

1. Dedicated Collection Phone: Use a clean smartphone reserved exclusively for testing.
2. Download App: Install Sensor Logger (iOS / Android).
3. Lock App Settings:
 - Sample Rate: Set to 50 Hz (20ms intervals).
 - Active Sensors: Toggle ON Accelerometer and Gyroscope.
 - Disabled Sensors: Toggle OFF Magnetometer, GPS, Audio, Orientation, Barometer.
 - Output Format: CSV.
4. Physical Station Location: Identify a quiet school stairwell connected to a 50-foot flat hallway.

Phase 2: Anonymous Master Ledger

Create a Google Sheet named KineTrace_Subject_Ledger.csv. Do not record participant names. Here is an example:

id	age	sex	height_cm	weight_kg	preexisting_factors	condition_category
sub_01	17	M	178	70	None	none
sub_02	18	F	165	58	Right ACL repair (2024)	joint_injury
sub_03	17	M	172	65	Flat feet (plantar fasciitis)	gait_structural

Condition Categories:

- none: Healthy, no reported conditions or recent injuries.
- joint_injury: Ligament tears (ACL/MCL), meniscus, sprains, or dislocations.
- gait_structural: Scoliosis, flat feet, leg-length discrepancy.
- chronic_neurological: Tremors, balance disorders, inner ear issues.
- active_pain: Current muscle soreness, active limp, or minor strain.

Phase 3: The 3.5-Minute Assembly Line Protocol

1. Assign anonymous ID (e.g., sub_01).
2. Ask the standardized screening question:
3. Ask: "Do you have any current or past injuries, surgeries, joint conditions, or structural factors (like flat feet) that affect how you walk, run, or stand?"

4. Log age, gender, height_cm, weight_kg, preexisting_conditions, and condition_category in your ledger.
5. Hand the phone to the subject and instruct them: "Place this in your front right pants pocket, top of the phone pointing down, screen facing your thigh."
6. Perform the following activities:
 - Sitting (sit): Participant sits still in a chair. Press Start Recording in Sensor Logger. Wait 30 seconds. Press Stop Recording. Save export as sub_01_sit.csv.
 - Standing (std): Participant stands completely still. Press Start Recording. Wait 30 seconds. Press Stop Recording. Save export as sub_01_std.csv.
 - Walking (wlk): Participant walks down the hallway at a natural pace. Press Start Recording. Walk for 60 seconds. Press Stop Recording. Save export as sub_01_wlk.csv.
 - Jogging (jog): Participant light-jogs down the hallway or track. Press Start Recording. Jog for 30 seconds. Press Stop Recording. Save export as sub_01_jog.csv.
 - Upstairs (ups): Participant stands at the bottom of the stairwell. Press Start Recording. Participant walks up stairs for 30 seconds. Press Stop Recording. Save export as sub_01_ups.csv.
 - Downstairs (dws): Participant stands at the top of the stairwell. Press Start Recording. Participant walks down stairs for 30 seconds. Press Stop Recording. Save export as sub_01_dws.csv.

Phase 4: Nightly Batch ETL Pipeline Script

Use process_community_data.py to transform raw Sensor Logger exports into the exact kinetrace_data_template.csv schema. Then analyze the data and test it with the KineTrace web app.